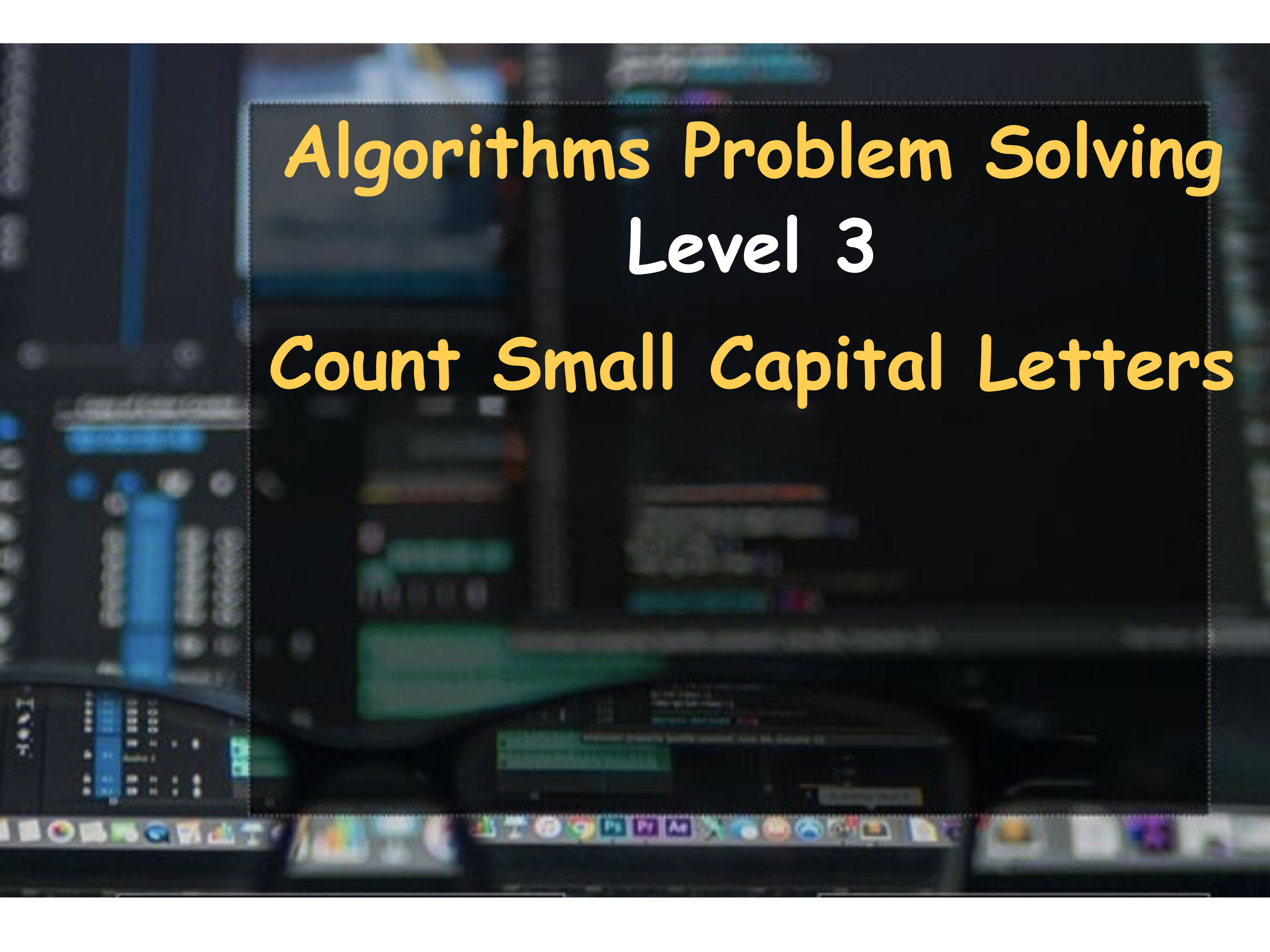
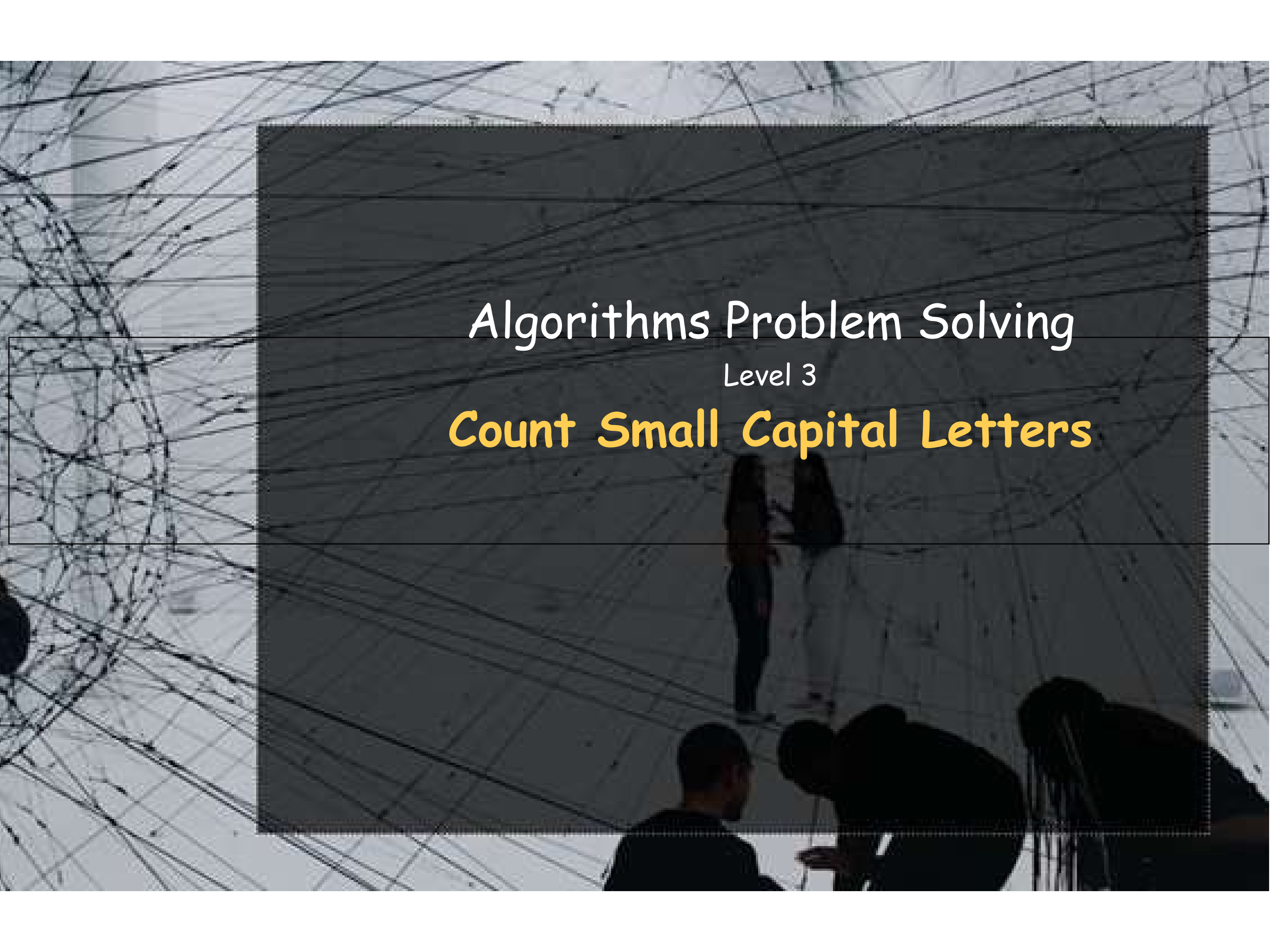




Algorithms Problem Solving Level 3

Count Small Capital Letters



Algorithms Problem Solving

Level 3

Count Small Capital Letters

Definition

Character Case Statistics

-  Program analyzes a string and counts uppercase letters, lowercase letters, and total characters.

-  It demonstrates character classification using standard `<cctype>` functions.

Explanation

Uppercase and Lowercase Classification

-  Each character is tested using ``isupper()`` and ``islower()``.

-  Separate counters are incremented according to the character category.

Total Length

-  ``str.length()`` returns the total number of characters in the string.

-  This includes spaces, digits, punctuation, and other characters.

Category Difference

-  The uppercase and lowercase counts do not necessarily add up to the total string length.

-  Characters that are neither uppercase nor lowercase account for the difference.

Procedures

1. Read the complete string.
2. Initialize uppercase and lowercase counters to `0`.
3. Traverse the string.
4. Count uppercase letters.
5. Count lowercase letters.
6. Obtain the total string length.
7. Display all statistics.

💡 Example

Input:

Sara Alterkawi

➡ Result

String Length = 14

Capital Letters Count = 2

Small Letters Count = 12

Note

- ◆ Spaces are included in `str.length()` but are not counted as letters.
- ◆ Digits and punctuation are also included in the total length but not in upper/lowercase counts.
- ◆ The same traversal can be extended to count digits, spaces, or punctuation.

⊘ Warning

- ◆ Do not assume `capitalCount + smallCount == str.length()`.
- ◆ The input must be read as a complete string when spaces need to be included.
- ◆ Character classification should use the appropriate `<cctype>` functions.



Comparison

▨ `CountLetters()` with Modes

- ✓ Can use an enum or parameter to select which character category to count.
- ✓ Centralizes the traversal logic.
- ✓ Reduces duplicated code.

▤ Separate Counter Functions

- ✓ Uses independent functions for uppercase, lowercase, and other categories.
- ✓ Can be easier to understand for very small programs.
- ✓ May duplicate string traversal logic.

✦✦ Summary

- ✓ Traverses a complete string character by character.
- ✓ Counts uppercase letters.
- ✓ Counts lowercase letters.
- ✓ Reports the total string length independently of letter counts.

Thank you 😊